

CSMFAB000000000040 V2.0

Segmented “Umbrella” Smart Rope Actuator — Production Iteration 2

Version 2.0 — Revised & Expanded | June 2026

Note: This document is the production refinement of CSMFAB000000000022. All V2.0 improvements from 022 apply plus the following production-specific updates:

Δ Change Log V1.0 □ V2.0 (Production-Specific)

- Segment count per metro: updated to nominal 4.2 segments/m (vs 4.0/m in 022)
- Quality control protocol: LIBS auto-analysis per segment batch (from Aegis-Forge I integration)
- Assembly line integration: segments produced by Aegis-Forge I downstream Machine 114
- Unit cost reduction: ZrB2 from in-house SHS synthesis = \$102/kg vs \$280 commercial
- Production lot certification: adapted from CSMFAB000000000001 Aegis-C quality protocol
- Segment connector geometry: updated to bayonet-lock (vs threaded in 022)

1. Production Design Differences from CSMFAB000000000022

1.1 Bayonet-Lock Connector (Production V2.0)

V1.0/022 used threaded PEEK connections: 5 turns to assemble each segment pair. At 1,600 segments per rope: 8,000 thread engagements □ 4-5 hours assembly time.

V2.0 Bayonet Lock: - Quarter-turn ZrO2 ceramic bayonet: 0.25 second per connection - Load rating: 2,800 N tensile (same as threaded equivalent) - Total assembly time 1,600 segments: ~7 minutes (automated assembly arm) - Mold NRE: \$8,500 for bayonet tool set

1.2 Aegis-Forge I Integration

Segments now produced as part of the Aegis-Forge ceramic output stream: 1. Aegis-Forge I Stage 8: processes zircon sand □ ZrO2 concentrate (S-11) 2. Phoenix Protocol converts S-11 □ ZrB2 powder via carbothermal route (48 hours) 3. Flash Sintering: ZrB2 powder □ sintered segment blanks 4. CNC grinding: blanks to 25.00±0.02 mm tolerance 5. LIBS spot-check: each batch of 100 segments analyzed for ZrB2 phase purity 6. Bayonet socket machining: Ti3AlC2 MAX Phase hinge installation 7. KNbO3-BaTiO3 actuator bonding (UV epoxy cure, 3 min) 8. LoRa module press-fit 9. Final test: segment deploys/retracts 10× without failure

1.3 Updated Performance Specifications

Parameter	V1.0	V2.0 Prod	Delta
Segment mass	9.2 g	8.6 g	-6.5% (bayonet lighter than thread)
Deployment time	1.2 ms	0.8 ms	-33% (MAX Phase hinge faster)
Tensile proof load	2,500 N	2,800 N	+12%
Operating temp range	-50°C to +200°C	-60°C to +400°C	Extended for re-entry
EMI SE	45 dB (Cu mesh)	92 dB (MXene)	+47 dB
Cost per segment (volume)	\$18	\$9.50	-47% (in-house ZrB2)

End of CSMFAB000000000040 V2.0 | Carrington Storm Motors